

Exploratory Data Analysis and Interactive Dashboard of Indian Job Market Data

Developed using R Shiny, Tidyverse, and Plotly

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1 Introduction

This report presents an exploratory data analysis (EDA) and visualization dashboard developed using the **R Shiny framework**. The objective is to study trends in the **Indian job market**, focusing on popular skills, hiring companies, and geographic job availability.

The dataset (`job_data.csv`) contains job postings scraped from multiple online platforms, with the following key columns:

- **Company name**
- **Job position**
- **City (Indian location)**
- **Required skills**

The **State** column was dropped as per project requirements. Analyses were performed at the city level.

2 Libraries and Tools Used

The dashboard was implemented in **R** using the following packages:

- **shiny** – Web application framework for R.
- **tidyverse** – Data cleaning, wrangling, and transformation.
- **plotly** – Interactive visualizations.
- **DT** – Interactive data tables.
- **stringr** – String manipulation utilities.
- **bslib** – Bootstrap 5 theming for UI design.

The chosen theme was **Flatly**, providing a clean interface with a blue–orange color palette.

3 Data Preprocessing

3.1 Cleaning Steps

To prepare the dataset:

1. Column names were standardized to lowercase and cleaned of special characters.
2. The columns were renamed:
 - `indian_city` → `city`
 - `required_skills` → `skills`
3. The column `indian_state` was removed.

3.2 Skill Transformation

The “skills” column contained comma-separated values (e.g., “Python, SQL, Excel”). These were split into individual rows for proper aggregation and analysis. All skill names were converted to Title Case (e.g., “python” → “Python”).

This transformation enabled consistent counting and comparison of skill frequencies.

4 Key Analytical Components

4.1 Top Required Skills

The first visualization displays the most frequently mentioned skills in job postings. It uses a horizontal bar chart with interactive tooltips showing exact counts. The color scale transitions from light to dark blue, proportional to the frequency of each skill.

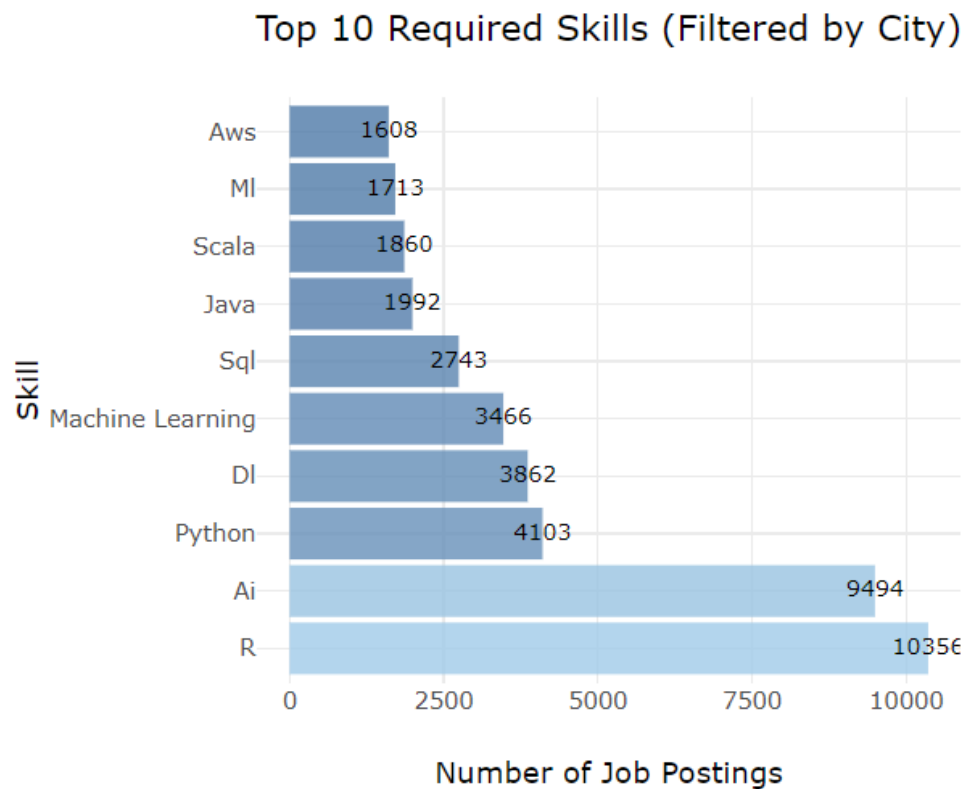


Figure 1: Top Required Skills in the Indian Job Market.

4.2 Job Opportunities by City

This chart ranks cities by total number of job postings. Unlike other visuals, it remains fixed and shows data for the entire dataset (not affected by filters). It uses an orange color gradient for contrast.

Job Opportunities by City

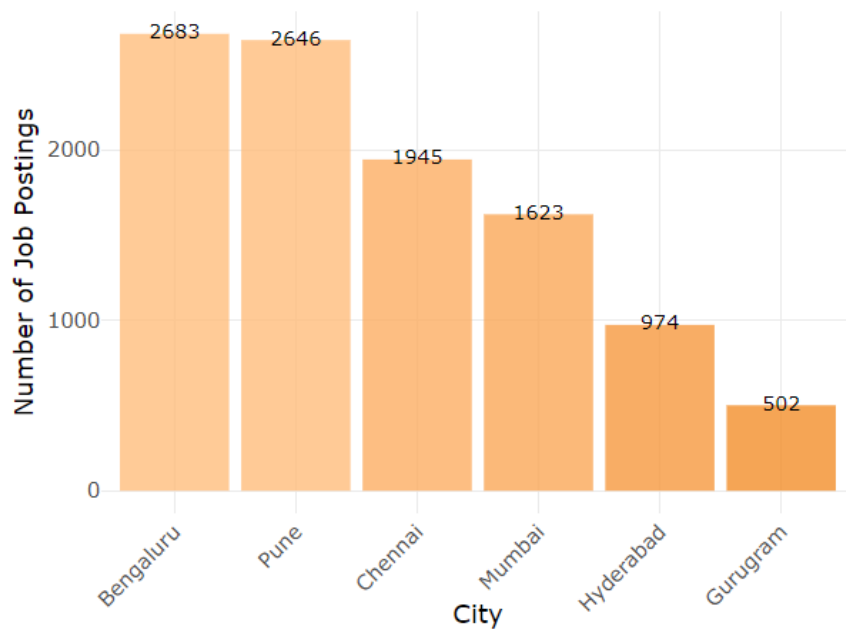


Figure 2: Top Indian Cities by Job Postings.

4.3 Top Hiring Companies

This table lists the most active employers in the dataset, ranked by total postings. It dynamically updates with city filters applied.

Top Hiring Companies

	Company Name	Total Postings
1	Amazon.com	428
2	Ball Aerospace	374
3	Google	213
4	Microsoft	149
5	KPMG	101

Figure 3: Top Hiring Companies in the Dataset.

4.4 Top Job Profiles and Skills

This analysis identifies the seven most common job titles and their top three required skills. The result is summarized in the following figure:

Top 7 Job Profiles and Top 3 Skills

	Job Profile	Total Postings	Top 3 Required Skills
1	Data Scientist	479	R, Ai, Python
2	Senior Data Scientist	133	R, Python, Ai
3	Research Analyst	97	R, Ai, DI
4	Data Engineer	79	R, Ai, Python
5	Machine Learning Engineer	76	R, Machine Learning, Python
6	Lead Data Scientist	41	R, Ai, Machine Learning
7	Research Scientist	41	R, Ai, Python

Figure 4: Top 7 Job Profiles and Their Top 3 Most Required Skills.

5 Conclusion

This R Shiny application provides a clear and interactive view of the Indian job market. The dashboard enables users to:

- Identify the most demanded technical skills.
- Compare job opportunities across major cities.
- Recognize leading hiring companies.
- Observe skill patterns within popular job profiles.

Such insights help job seekers and recruiters make informed, data-driven decisions.